

Navigating the Human-oversight Dilemma in AI-based Systems

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Abstract— As artificial intelligence (AI) systems increasingly assume responsibility for critical decision-making, organizations face the challenge of balancing full automation with appropriate human oversight. While automation offers notable gains in efficiency, scalability, and performance, it also raises concerns regarding safety and ethical integrity. Overdependence on AI may result in biased outcomes, diminished transparency, and erosion of user trust, whereas excessive human intervention can compromise system efficiency and usability. In this research in progress, we investigate the inherent trade-offs between full automation and human oversight by analyzing the tensions that arise within AI decision-making processes. Utilizing an ontology-based framework, we systematically address these trade-offs and explore conditions under which human intervention is most beneficial. Drawing on a multi-criteria decision analysis (MCDA) approach, we propose methods for optimizing human oversight without undermining the strengths of AI. By doing so, this study offers advances for the development of trustworthy AI systems that uphold regulatory compliance and ethical responsibility, while maintaining operational effectiveness. Future research is needed for evaluating the effectiveness of the proposed methods when implemented in the field.

Keywords—RE, RE4AI, trustworthiness, questionnaire, ontology

I. INTRODUCTION

The rapid integration of artificial intelligence (AI) across domains such as healthcare, finance, and transportation has shifted complex decision-making from humans to machines. While this boosts efficiency, it also raises pressing questions about human oversight: When should humans intervene, and how much oversight is optimal? Achieving the right balance is vital—too little oversight may lead to biased or opaque decisions whereas too much may reduce system performance and reintroduce human error. In this research in progress, we examine these dilemmas using an ontology-based framework for requirements engineering for AI (RE4AI). We propose a structured framework for managing trade-offs between automation and varying degrees of human oversight, while accounting for their impact on key system objectives and constraints such as explainability, ethical bias mitigation, and safety. The framework integrates:

- An ontology model mapping requirement interdependencies,
- A stakeholder-driven questionnaire to uncover conflicts, and

- Negotiation tools for resolving trade-offs.

The research question is defined as follow:

RQ: How can requirements engineers of AI-powered systems determine the appropriate balance of human oversight and automation to ensure system trustworthiness, while keeping it reasonably efficient and scalable?

This study builds on our prior RE4AI ontology work, which identified key contradictions surrounding human oversight in AI-powered systems [1]. Here, we deepen the analysis and introduce a systematic framework to resolve these tensions through trade-off balancing. Our key contributions include: (1) a conceptual framework explicitly designed to balance oversight and automation, (2) integration of real-world dilemmas from multiple domains, and (3) grounding in stakeholder perspectives using an ontology-based interview tool. To our knowledge, no existing RE methodology systematically addresses the human-oversight dilemma in early system design. Adopting a mixed qualitative approach, we conducted a literature review, analyzed real-world domain-specific cases reported in the literature, and conducted 35 interviews with stakeholders involved in AI-based system development, to inform the design of the framework.

The rest of the paper is organized as follows. Section II presents our review and analysis of the literature related to the trade-offs between human oversight and automation in AI. Section III brings insights from a literature-based case analysis, and Section IV reports on the interviews with stakeholders. Section V details our proposed framework. Section VI provides an illustrative example of using the framework. Section VII presents our plans for future research and Section VIII concludes.

II. CONTRADICTIONS AND TRADE-OFFS IN HUMAN-AI INTERACTION

Our earlier work introduced an ontology for requirements engineering (RE) in AI-based systems, mapping requirement categories and highlighting key contradictions, in particular, the tension between human oversight and other system goals [1]. Amershi et al. [2] proposed 18 design guidelines for human-AI interaction across the user journey (before, during, after, and over time), emphasizing trust, usability, and transparency. However, such guidelines frequently overlook critical trade-offs with other requirements like accuracy, efficiency, ethics, and security. Our framework addresses this gap by

systematically identifying and negotiating these competing demands. Figure 1 presents a conceptual model that illustrates the inherent conflicts between human-AI interaction and other system requirements. The contribution of this model lies in supporting system designers in identifying and resolving conflicts that arise during the RE process. Managing trade-offs through structured negotiation tools constitutes a key phase within the developed framework.

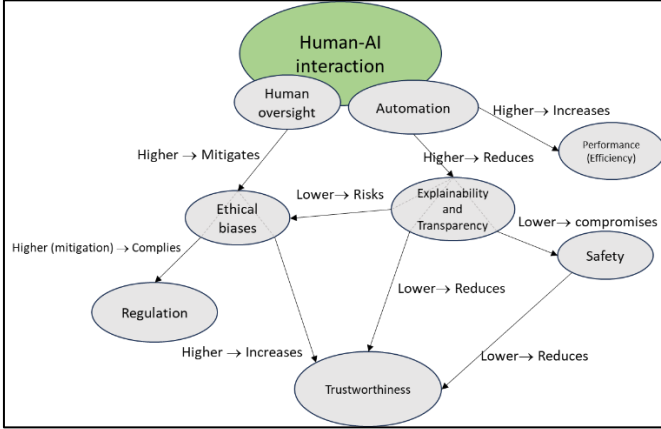


Figure 1 - Conceptual model of conflicting requirements in relation to human-AI interaction

Trustworthiness: A core tension in AI design exists between automation and system trust. As AI systems become more autonomous to improve performance, they often sacrifice transparency and explainability, key components of user trust [3, 4, 5]. Users may struggle to understand or challenge AI outputs, especially when models are opaque. Conversely, increasing human oversight can reduce system efficiency and reintroduce human error. Balancing trust requires defining human roles and tailoring explainability to user needs.

Ethics: Reducing human involvement may improve efficiency but risks undermining fairness, especially when biased data drives opaque models [3, 6]. Mitigating bias usually requires human oversight, which can slow decision-making and reduce consistency [7, 8]. Similarly, increasing transparency to enable oversight may conflict with privacy, as explainability often requires exposure of sensitive data. Addressing these contradictions calls for context-aware strategies, such as selective transparency and privacy-preserving techniques (e.g., differential privacy, federated learning), alongside dynamic bias mitigation mechanisms.

Performance: High-performance AI models frequently operate as “black boxes,” prioritizing speed and accuracy over explainability. Requiring human oversight or explainability can limit model complexity, reduce predictive power, and add computational overhead [8, 9]. Furthermore, pausing for human validation introduces latency and variability [10]. In domains like healthcare, explainability may justify performance trade-offs, whereas in cybersecurity or mobility, real-time response is critical. Thus, performance-oversight trade-offs must be context-specific.

Safety: While human oversight is typically viewed as a safety mechanism, especially in domains like healthcare and mobility, it can introduce delays and inconsistencies in real-time systems [3, 4, 8]. In some cases, safety stems from how the system is built and tested, not from having a human involved. Moreover, perceived safety based on human involvement may not align with actual safety. Resolving this contradiction requires aligning oversight levels with risk, combining automation with auditable behavior, and supporting regulatory clarity and stakeholder trust [10, 11].

Explainability & Transparency: Human oversight depends on understanding AI outputs, but explainability often conflicts with system performance. Simpler, interpretable models sacrifice accuracy, while deep learning systems are opaque [6, 12]. Post-hoc explanations offer limited fidelity and may mispresent system behavior. Different stakeholders also require different explanation types, which complicates system design. Additionally, explainability features add computational load and may reduce throughput [8, 9, 13].

This literature review underscores the complex interdependent tensions between human oversight and other critical requirements in AI-based systems, including trust, ethics, performance, safety, and explainability. While prior works offer foundational frameworks, they often fail to systematically address the contradictions that emerge when these requirements conflict. Our proposed framework fills this gap by explicitly mapping these trade-offs and fostering adaptive approaches aligned with system objectives. The literature review reveals that human oversight, while central to trust and ethical governance, may diminish system efficiency, compromise performance, and introduce new risks. Conversely, reducing oversight to enhance performance can undermine fairness, transparency, and user trust. These findings call for a nuanced, application-specific calibration of oversight mechanisms, balancing automation with explainability, performance with responsibility, and user empowerment with systemic robustness.

III. LESSONS FROM LITERATURE-BASED CASE ANALYSIS

To contextualize the conflicting system objectives identified in the initial literature review, we analyzed a diverse set of real-world case studies drawn from both academic literature and formal media sources. These examples highlight recurring conflicts between human oversight and competing system goals. Given the limited access to AI solutions in proprietary systems, literature-based and media-documented cases offer valuable insights into the practical challenges encountered during the implementation of these AI solutions. Examining reported real-life cases enables us to examine how different domains implement, adapt, or struggle with human-oversight mechanisms. The insights derived from these cases inform and refine our proposed framework, offering concrete illustrations of oversight dilemmas and potential resolution strategies.

Autonomous Vehicles: Self-driving technologies developed by Tesla and Uber demonstrate the risk of full automation in safety-critical environments. Accidents involving autonomous vehicles have underscored the need for human-in-the-loop

mechanisms, such as driver fallback systems or remote human intervention, to handle complex, unpredictable conditions [14, 15, 16].

Fraud Detection in Banking: AI systems effectively detect fraudulent activity, but may trigger false positives, leading to legitimate transactions being blocked. In these cases, human analysts are needed to verify and override AI decisions, ensuring efficiency without compromising user experience [17, 18, 19].

Medical Diagnostics: AI models have outperformed doctors in several specific tasks, such as cancer detection, but their black-box nature undermines trust. Clinicians require explainable outputs before acting on AI recommendations. This requirement was addressed by integrating explainable AI (XAI) tools, like saliency maps, but these often impose performance costs and fall short of full transparency [13, 20, 21].

AI in Hiring: Many companies including, for example, Amazon and LinkedIn, use AI for resume screening and behavior prediction. However, the AI models used for this purpose have limited transparency, raising fairness and accountability concerns. Organizations are therefore forced to choose between simplifying models for explainability or providing post-hoc rationalizations with limited fidelity [22, 23, 24].

Mortgage Approvals: Fairness-efficiency trade-offs are apparent in AI-assisted lending. Implementing fairness measures, such as demographic parity or equal opportunity, can help reduce systemic biases and promote equitable access to credit, especially for historically disadvantaged groups. However, such measures may lead to the approval of applicants who do not meet financial risk criteria, potentially increasing default rates and resulting in financial losses for lenders. On the other hand, when lenders adjust AI models to satisfy fairness metrics (e.g., equal approval rates across demographic groups), they may reject applicants who are financially qualified but fall outside the fairness criteria. This highlights the tension between fairness goals and financial optimization [25].

Predictive Policing: Policing systems, such as PredPol (Predictive Policing), identify high-risk areas using historical crime data. These tools can reinforce racial biases and lead to over-policing. Cities governmental authorities have responded with human oversight mechanisms and community input, but this measure slows AI responsiveness and adds operational complexity [26, 27, 28].

AI in Legal Reasoning: Explainable AI is highly important in law, where accountability and reasoning are foundational. Decision-support tools based on symbolic AI (e.g., Abstract Dialectical Frameworks) help legal professionals interpret AI recommendations. However, these tools frequently require significant human input and are computationally expensive, limiting scalability [29].

AI for Pandemic Triage: During the COVID-19 pandemic, hospitals deployed AI models to support triage and resource allocation. Although these tools improved efficiency, most of them were trained on narrow datasets and risked biased outcomes. Experts emphasized the need for clinician oversight,

contextual judgment, and ethical review to ensure fairness and accountability [30].

Aerospace Predictive Maintenance: AI enhances operational efficiency through predictive maintenance, reducing downtime and costs. However, human validation remains critical. Technicians reviewing AI-generated alerts reduce false positives and ensure safety in high-risk environments, reinforcing the value of structured oversight [16, 31].

Across sectors, common contradictions emerge: high-performance models struggle with explainability; fairness often demands manual intervention; and real-time systems face trade-offs between responsiveness and oversight. Human-in-the-loop mechanisms improve trust and accountability but introduce operational costs and design complexity, and at times potential human errors. These case studies reaffirm the need for domain-specific, ethically grounded oversight strategies that support both technical performance and legitimacy.

IV. STAKEHOLDERS' INTERVIEWS

The ontology-based questionnaire, which was developed in our previous works [1], was used to conduct semi-structured interviews with stakeholders from different domains (e.g., defense, transportation, finance, cyber security, etc.) involved in the development of AI-based systems. The interview study was designed to capture diverse perspectives from professionals involved in AI-powered system. A total of 35 stakeholders, from a wide range of professional backgrounds and expertise, participated in the interviews. The vast majority of participants (32/35) held senior positions, including project managers, system engineers, technology managers, AI experts, and department heads. Their professional experience was notably extensive, with 27 of the 35 participants having over a decade of experience, and 10 exceeding 15 years in their fields.

While the sample of participants was primarily composed of experts from the defense sector, it also included participants from other key domains such as transportation, cybersecurity, finance, automotive, governance, retail-tech, and the judicial system. The extensive representation from the defense sector provided unique value to this study, given its focus on high-stakes, mission-critical systems where human oversight, safety, and accountability are paramount. The defense domain's rich experience in managing complex, high-risk systems offers important lessons and best practices that are increasingly relevant as AI systems become progressively more embedded in various critical sectors.

The semi-structured interview protocol was based on a questionnaire structured around our developed RE ontology and addressed a broad range of requirement categories. For this study, we focused on a selected subset of the full questionnaire, concentrating on human-AI interaction. Accordingly, the interview protocol included the following questions:

- 1 – *What is the anticipated level of human involvement in the system operation?*
- 2 – *What are the primary justifications for human intervention within the system operation?*

- 3 – *How should the system balance automation and user control to maintain usability?*
- 4 – *How can we provide human oversight while maintaining system efficiency?*
- 5 – *What ethical considerations should be accounted for in human-system interactions?*
- 6 – *How does reducing human oversight impact user trust, and in which scenarios should human intervention be mandatory to maintain trust?*

To explore factors shaping appropriate user oversight and its alignment with broader system requirements, we applied thematic analysis to the data elicited via the stakeholder interviews [32]. The thematic analysis was conducted using an inductive approach. The first author manually reviewed all interview transcripts, systematically identifying recurring and salient themes that emerged from the participants' responses. The analysis followed an iterative process of reading, coding, and grouping similar statements to uncover patterns in the data. While the analysis was primarily inductive, it was also informed by the themes identified in the prior literature review, allowing for triangulation between existing theoretical concepts and the empirical findings [33]. This dual focus enabled the identification of both pre-existing themes, which validated the literature-based insights, and new emergent themes that reflected participants' unique perspectives on human oversight. The analysis yielded the identification of six key themes that highlight trade-offs and design considerations essential for implementing effective human oversight, all of which should be addressed throughout the RE process.

Theme 1: Trust and Perceived Control. Participants emphasized that user trust often hinges on a sense of control, even if actual autonomy is limited. Virtual decision points, where the user is given the opportunity to review, confirm, or override the system's recommendations, enhance perceived influence, particularly in high-risk domains like military or legal systems. Several interviewees proposed positioning AI as a recommendation engine rather than an autonomous decision-maker to preserve user agency.

Theme 2: Explainability. Explainability is a requirement shared by several types of stakeholders, but its form must be audience specific. Developers require technical diagnostics, while end-users need intuitive, high-level narratives. In regulated sectors (e.g., law, healthcare), explainability is essential for compliance and ethical assurance.

Theme 3: Oversight modes and human-in-the-loop design. The participants referred to three oversight modes:

- Human-in-the-loop, refers to scenarios where humans are actively involved in decision-making and must approve or intervene before the system proceeds.
- Human-on-the-loop, describes settings in which the AI operates autonomously but is continuously monitored by a human who can intervene if necessary.
- Human-out-of-the-loop applies when the system operates independently without real-time human oversight.

The closer the user is to the loop, the more explainability is required. Critical decisions demand clearly defined human roles and checkpoints.

Theme 4: Oversight in real-time systems. Real-time systems (e.g., autonomous vehicles, security applications) present unique constraints on oversight due to latency demands. Human feedback must be selective, fast, and context aware. Oversight in these systems must balance responsiveness with cognitive load.

Theme 5: Stakeholder-centric interaction design. Oversight must reflect stakeholders' roles and needs. Different user types, legal experts, operators, and data scientists require different interfaces, access levels, and explanation styles. Co-design with stakeholders was proposed as a strategy to align system workflows with real-world decision-making.

Theme 6: Monitoring, feedback, and learning from users. Continuous monitoring of user behavior and system impact is essential for refining human-AI interaction. Systems should learn from feedback to adapt over time and foster trust. Monitoring is also needed to detect misuse or misalignment with expectations.

Stakeholders consistently highlighted the importance of aligning oversight mechanisms with user expectations, domain constraints, and operational contexts. Trust-building features, such as override capabilities and tailored explainability, needs to be embedded already in the design phase. In fast-paced environments, oversight needs to be agile and non-disruptive. Importantly, stakeholder involvement must continue post-deployment, through monitoring tools and adaptive feedback loops. Yet, these user-centered requirements often conflict with other system demands, such as performance, efficiency, or privacy. Addressing them calls for a balancing process, one that integrates stakeholder needs without undermining technical goals.

V. FRAMEWORK FOR OVERSIGHT BALANCING

To navigate the tension between human oversight and competing system requirements, we propose a structured, context-sensitive framework for determining appropriate levels of human involvement in AI-based systems.

Foundational theories in human factors guide the design of effective human-automation oversight, even before AI, in its current form, was introduced in the market. Rasmussen's Skill-Rule-Knowledge (S-R-K) model categorizes human behavior into automatic (skill-based), procedural (rule-based), and deliberative (knowledge-based) levels, highlighting the need for oversight mechanisms that adapt to the user's cognitive mode and task complexity [34]. Similarly, Endsley's Situational Awareness (SA) model outlines three levels, perception, comprehension, and projection, emphasizing the importance of system support for accurate understanding and anticipation in dynamic environments [35]. Together, these models inform our framework by underscoring the need to align system autonomy and user control with cognitive demands and contextual awareness.

Several prominent studies offer significant guidelines for effective human oversight in AI systems. The RE4HCAI

framework emphasizes embedding human oversight through clear architectural modes, human-in-the-loop, human-on-the-loop, and human-in-command, while highlighting the importance of explainability, responsibility, and early integration of oversight mechanisms during system design [REF]. The EU AI Act establishes legal requirements for human oversight, focusing on minimizing risks in critical-safety domains, and fundamental rights. It mandates technical and organizational measures to ensure human operators can understand system limitations, intervene when necessary, and take ultimate control. The study of Sterz, Sarah, et al. [REF] proposes a structured, interdisciplinary model for effective oversight, centering on four key conditions: causal power (ability to intervene), epistemic access (system understanding), self-control (attentiveness and capacity to act), and fitting intentions (motivation and responsibility). It further identifies practical enablers and barriers across technical, individual, and organizational levels. Together, these studies highlight the need for human oversight to be designed systematically, blending technical features, legal safeguards, and human factors to ensure accountability, operational effectiveness, and risk mitigation in AI-powered systems.

Insights from prior works on human oversight, including risk-mitigation, user perception aspects and regulatory guidelines, were carefully embedded within the framework and are reflected in its structure a. The framework builds upon these foundations and adds aspects by integrating human oversight considerations directly into the RE process. The additional aspects embedded in the framework include an iterative, lifecycle-oriented approach, where oversight is continuously refined through stakeholder engagement by on-going evaluation, and operational feedback.

A key contribution of this framework is its mapping of human oversight to an RE ontology, enabling a systematic and structured identification of oversight requirements alongside other system requirements. By doing this, the framework addresses trade-offs between human oversight and other critical requirements such as performance, efficiency, and usability, broadening the conventional focus related to risk and ethics. This holistic and practical approach allows for more balanced and context-sensitive oversight strategies throughout the AI-powered system’s RE.

The framework processes begin with defining the operational context, including domain-specific regulations, ethical considerations, and risk profiles. This step identifies potential harms, ethical constraints (e.g., fairness, privacy), and the possible impact of system decisions. Decisions with significant consequences typically necessitate greater levels of human involvement.

Figure 1 describes the developed oversight balancing framework.

The framework builds on the oversight spectrum identified through both the literature review [10, 36], and insights shared by the interviews' participants:

- Human-in-the-loop (HITL): Humans approve decisions before execution.

- Human-on-the-loop (HOTL): Humans monitor and may intervene in real time.
- Human-out-of-the-loop (HOOTL): AI operates autonomously.

Three key oversight mechanisms are emphasized:

- Explainability: Enables users to interpret and audit AI decisions.
- User control and override capabilities: Enables user intervention when disagreeing with the AI's decision.
- Monitoring and feedback loops: Track system performance and user interactions to detect anomalies and support adaptation.

Legend:

Rectangles – Decision points: A key decision (D) or activity (A) that must be addressed at that phase of the oversight framework.

Straight arrows – Sequential flow: Indicate the logical progression in the oversight design process.

Curved arrow – Iterative feedback loop: Highlight stages where evaluation results may lead to adjustments in earlier phases, supporting continuous refinement.

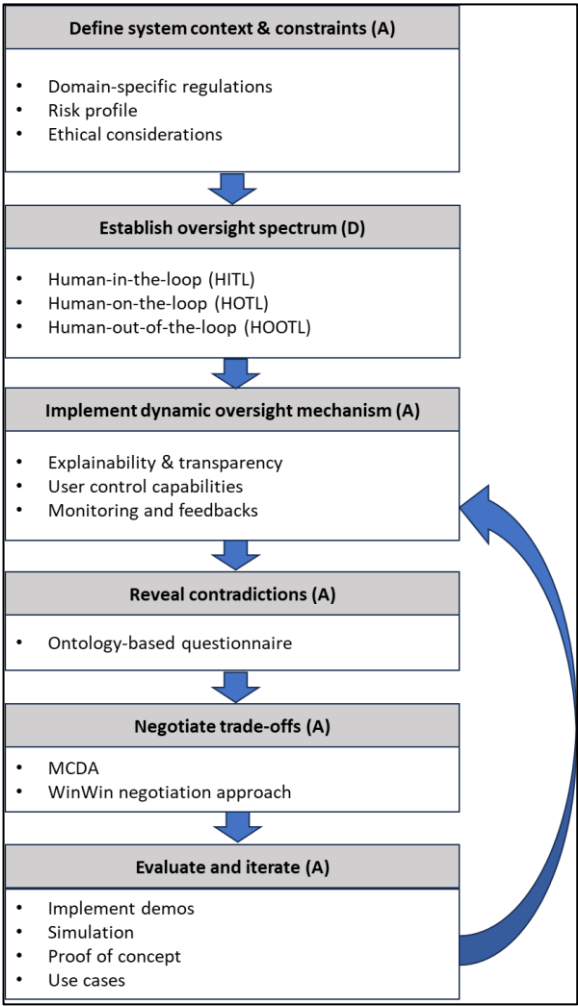


Figure 1 – Oversight balancing framework

Oversight is treated as dynamic and iterative. Tools such as simulations, demonstrations, and proof-of-concept (PoC)

evaluations allow developers and other stakeholders to test and refine oversight strategies in diverse contexts. This iterative process helps address emerging risks, evolving expectations, and domain-specific constraints. To manage conflicts between oversight and other system objectives, the framework guides the use of structured negotiation tools.

Conflicts negotiations are facilitated through the use of the following tools:

- Ontology-based questionnaire [1]: Our structured questionnaire plays a key role in revealing hidden contradictions between requirements, by actively involving key stakeholders. Capturing diverse viewpoints from engineers, business leaders, regulators, and end-users, surfaces conflicting priorities and exposes implicit biases in requirements formulation.
- Multi-Criteria Decision Analysis (MCDA): MCDA assists in providing a structured and transparent approach to evaluate and prioritize conflicting requirements [37].
- Win-win negotiation approach: a collaborative method developed to help stakeholders reach mutually satisfactory agreements during the early phases of system development. It is built on the principle that all key stakeholders should “win,” that is, have their main concerns and objectives addressed, for a successful and sustainable system design [38].

While the framework encourages the use of MCDA and win-win negotiation approach to support negotiation and trade-off resolution, it does not currently incorporate these methods directly nor does it provide additional implementation layers beyond abstract guidance. These tools guide stakeholders in evaluating trade-offs and making transparent, participatory decisions that align technical feasibility, system objectives, and ethics regulations.

VI. ILLUSTRATIVE EXAMPLE: HUMAN OVERSIGHT IN AN AI-POWERED HOMELAND SECURITY SYSTEM

To demonstrate the application of the framework, we present an illustrative example based on a homeland security system using an AI-based model for threat detection. The system was designed to automatically identify potential threats and suggest interception actions in real time.

During the requirements elicitation process, two key conflicting requirements were identified:

- Requirement A: Maximize automation to enable rapid threat detection and response, ensuring minimal reaction time under high-pressure conditions.
- Requirement B: Maintain human oversight to allow operators to review and approve system-recommended actions before engagement, ensuring accountability and reducing the risk of false alarms.

Applying the framework’s trade-off analysis revealed that different users with different objectives, each operating under distinct operational doctrines, had varying oversight needs. One objective prioritized speed and operational performance, preferring higher automation with minimal human intervention. The other placed greater emphasis on control and accountability, even if it meant accepting slower system responses to allow for thorough human review before action.

By using the framework’s negotiation tools, the development team was able to map these trade-offs systematically and adapt the system’s oversight configurations to the preferences of the different user types. This example highlights the framework’s capability to navigate complex oversight-performance tensions and tailor oversight mechanisms to diverse operational contexts.

VII. FUTURE WORK

As part of our planned future work, we intend to validate and potentially refine the proposed framework through additional experts specializing in human-system interaction. In these sessions, we will present the framework and invite experts to critically assess its structure, components, and applicability, as well as to suggest possible improvements. To support focused reflection, we have prepared a set of guiding questions aimed at eliciting detailed qualitative feedback:

- 1 – *How effectively does the initial phase of the framework support capturing real-world operational context and user diversity? What additional contextual elements (e.g., cognitive workload, environmental stressors) should be considered?*
- 2 – *Does the framework adequately address all relevant human oversight dimensions (e.g., control, transparency, trust, intervention readiness)?*
- 3 – *What user-centered design principles should guide the development of oversight mechanisms?*
- 4 – *What evaluation methods (e.g., simulation, cognitive walkthroughs, usability testing) would be most effective for refining oversight mechanisms?*
- 5 – *Have you previously applied structured methods (e.g., MCDA, win-win models, stakeholder workshops) to resolve trade-offs in system design?*

In addition, we plan to conduct a quantitative validation by presenting a series of structured statements, derived from the framework, to a broader group of stakeholders from various domains. These stakeholders will be asked to express their level of agreement with each statement using a Likert scale. This combined approach, merging expert interviews with survey-based analysis, will enable us to evaluate the broader relevance and acceptance of the framework across diverse user groups and operational settings.

VIII. DISCUSSION

While the proposed oversight framework offers a structured, theory-based method to managing human oversight in AI systems, its broader applicability across diverse domains remains to be tested. An initial examination, using reported real-world cases focused on high-stakes, regulated sectors, including healthcare, finance, and law. However, as AI expands into fields like education, social media, and public services, each with distinct ethical and operational constraints, further evaluation is needed.

Importantly, the findings of the study, based on both literature and interview insights, indicate that resolving requirement contradictions does not require novel negotiation

tools. Rather, established methods like MCDA and the win-win approach, remain effective for structuring stakeholder trade-offs, provided they are tailored to AI-specific features such as model opacity, continuous learning, and shifting user trust. The framework itself does not embed these tools but rather recommends their integration to support early conflict identification and structured deliberation.

When considering the findings of this research in progress, several threats to validity need to be acknowledged, particularly given the interdisciplinary and evolving nature of AI systems and the complexity of designing oversight mechanisms.

Construct validity may be affected by differing stakeholder interpretations of key terms such as “human oversight,” “explainability,” and “trust.” While the ontology-based questionnaire and structured interviews aimed to standardize these concepts, variations may persist across contexts.

Internal validity is limited by the qualitative nature of the evidence, primarily expert interviews and case studies. We addressed these concerns by comparing findings with existing research and real-world examples. However, future studies using simulations or controlled tests are needed to better understand cause-and-effect relationships.

External validity may be constrained by the framework’s focus on high-stakes, regulated domains. While the design is intended to be scalable and domain-agnostic, broader validation across sectors such as education or services is necessary to confirm generalizability.

Finally, methodological integration remains a challenge. The framework links ontology modeling, qualitative insights, and negotiation tools, but these connections are primarily conceptual. Future work should formalize these linkages, for example, by showing how stakeholder input informs trade-off prioritization, to enhance coherence and usability.

In summary, while many of these limitations are addressed through structured methods and triangulation, further empirical testing, broader domain application, and clearer methodological alignment are needed to strengthen the framework’s validity and applicability.

IX. CONCLUSIONS

This research underscores the inherent tension between the efficiency of automation and the necessity of human oversight in AI-based systems. Oversight should not be treated as a binary feature but as a dynamic, context-sensitive design element shaped by domain, performance, and ethical constraints. The proposed framework addresses this complexity through a layered oversight spectrum, structured stakeholder engagement via an ontology-based questionnaire, and trade-off negotiation tools. A key insight is that human oversight must evolve over time. As systems mature and risks shift, oversight mechanisms should be iteratively refined through simulations, pilot applications, and real-world feedback to ensure continued alignment with human values and operational realities.

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